Advertising Campaign and Product Analysis using PostgreSQL

Tools Used: PostgreSQLSQL

Dataset Size: 846,811 records

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PROJECT OBJECTIVES

The objective of this project is to analyze advertising product data to understand how advertisers structure campaigns and promote products.

Key Questions

- Which product categories dominate advertising campaigns?
- Do campaigns advertise products across wide price ranges?
- Which Categories Have the Most Expensive Products?

DATASET OVERVIEW

The dataset contains advertising product information used in online campaigns.

Dataset Characteristics

- Total records: 846,811
- Advertising campaigns: multiple campaigns across categories
- Product pricing information included

Column	Description
adgroup_id	Unique advertisement
cate_id	Product category identifier
campaign_id	Advertising campaign
customer	Advertiser ID
brand	Brand identifier
price	Product price

SQL QUERIES

```sql
-- 4. Top product categories
SELECT cate_id,
COUNT(*) AS ad_count
FROM ad_features
GROUP BY cate_id
ORDER BY ad_count DESC
LIMIT 10;
```

	cate_id bigint	ad_count bigint
1	6261	42691
2	4520	21344
3	1665	15443
4	4282	14895
5	6426	14159
6	6423	14130
7	4505	13495
8	4280	11738
9	4284	11044
10	4283	10794

A small number of product categories dominate advertising activity, with category 6261 having the highest ad volume (42,691 ads).

SQL QUERIES

```sql
-- Do campaigns advertise products across wide price ranges?
SELECT campaign_id,
MIN(price) AS min_price,
MAX(price) AS max_price
FROM ad_features
GROUP BY campaign_id
ORDER BY max_price DESC
LIMIT 10;
```

	campaign_id bigint	min_price double precision	max_price double precision
1	44251	3000	100000000
2	218918	100000000	100000000
3	270719	1500	100000000
4	266086	88	100000000
5	270027	2300	100000000
6	218101	880	100000000
7	8401	1	88900000
8	37665	300000	15000000
9	417722	4500	10000000
10	416333	9900000	9900000

Several campaigns advertise products across extremely wide price ranges, indicating diverse product targeting within single campaigns

SQL QUERIES

```sql
--Which Campaigns Advertise the Most Expensive Products?
SELECT campaign_id,
AVG(price) AS avg_price
FROM ad_features
GROUP BY campaign_id
ORDER BY avg_price DESC
LIMIT 10;
```

	campaign_id bigint	avg_price double precision
1	218918	100000000
2	218101	11149420
3	270027	10022660
4	416333	9900000
5	37665	7650000
6	8401	6020972.791666667
7	239302	5888888
8	270719	5074824
9	132721	3980000
10	135906	3660000

Certain campaigns consistently promote high-value products, with campaign 218918 showing the highest average product price.

SQL QUERIES

```sql
--Price Tier Market Distribution
SELECT
CASE
WHEN price < 50 THEN 'Budget'
WHEN price BETWEEN 50 AND 200 THEN 'Mid-range'
WHEN price BETWEEN 200 AND 500 THEN 'Premium'
ELSE 'Luxury'
END AS price_segment,
COUNT(*) AS ads
FROM ad_features
GROUP BY price_segment
ORDER BY ads DESC;
```

	price_segment text	ads bigint
1	Mid-range	313405
2	Budget	214968
3	Premium	169273
4	Luxury	149165

Mid-range products dominate advertising, accounting for the largest share of ads (313,405), followed by budget products.

KEY INSIGHTS

Advertising activity is concentrated in a few product categories, indicating focused marketing efforts.

- Mid-range products dominate the advertising market, suggesting campaigns target mainstream consumers.
- Some campaigns promote significantly higher-priced products, indicating premium market targeting.
- Several campaigns advertise products across wide price ranges, reflecting diversified campaign strategies.
- Luxury products appear less frequently but represent a niche high-value advertising segment.

FUTURE IMPROVEMENTS

Possible Extensions :

- Integrate click data to measure Click-Through Rate (CTR)
- Perform campaign performance analysis
- Develop predictive models for advertisement engagement
- Build an interactive dashboard for marketing insights

Thank you